UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of Computing

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LAB ACTIVITY

SECP 2523 DATABASE (WBL)

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LAB 1 - Creating a Database

CREATING THE TENNIS LOGOS DATABASE

Tennis Logos (TL) supplies customized tennis clothing and accessories for clubs and tournaments.The company purchases these items from suppliers at wholesale prices and adds the customer's logo. The final item price is determined by marking up the wholesale price and adding a fee that is based on the complexity of the logo.

Tennis Logos also does graphic design work for customers. Currently, the information about the items and the suppliers is stored in two Excel workbooks. Each item is assigned to a single supplier, but each supplier may be assigned many items.

You are to create a database that will store the item and supplier information. You already have determined that you need two tables, a Supplier table and an Item table, in which to store the information.

Instructions: Perform the following tasks.

1. *Use the Blank desktop database option to create a new database in which to store all objects related to the items for sale. Call the database – TL_Sport Database.*

Answer:

```
mysql> CREATE DATABASE TL_Sport ;
Query OK, 1 row affected (0.00 sec)

mysql> USE TL_Sport ;
Database changed
```

2. *Create the Supplier table using the structure shown in Figure 1. Use the name Supplier for the table.*

Answer:

```
mysql> CREATE TABLE Supplier(
    -> SupplierCode Char(2) PRIMARY KEY,
    -> SupplierName VARCHAR(30),
    -> TelephoneNumber VARCHAR(12)
    -> );
Query OK, 0 rows affected (0.02 sec)
```

3. *Add the data shown in Figure 2 to the Supplier table.*

Answer:

```
mysql> INSERT INTO Supplier (SupplierCode, SupplierName, TelephoneNumber)
    -> VALUES
    -> ('LM', 'Last Merchandisers', '803-555-7641'),
    -> ('PC', 'Pratt Clothing', '517-555-3853'),
    -> ('SD', 'Scryps Distributers', '610-555-8741') ;
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

4. *Print the Supplier table.*

Answer:

```
mysql> SELECT * FROM Supplier ;
+--------------+---------------------+-----------------+
| SupplierCode | SupplierName        | TelephoneNumber |
+--------------+---------------------+-----------------+
| LM           | Last Merchandisers  | 803-555-7641    |
| PC           | Pratt Clothing      | 517-555-3853    |
| SD           | Scryps Distributers | 610-555-8741    |
+--------------+---------------------+-----------------+
3 rows in set (0.00 sec)
```

5. *Create the Item table using the structure shown in Figure 3. Use the name Item for the table.*

Answer:

```
mysql> CREATE TABLE Item(
    -> ItemNumber CHAR(4) PRIMARY KEY,
    -> Description VARCHAR(20),
    -> OnHand INT ,
    -> WholesalePrice DECIMAL(5, 2),
    -> BaseCost DECIMAL(5, 2),
    -> SupplierCode CHAR(2),
    -> FOREIGN KEY(SupplierCode) REFERENCES Supplier(SupplierCode)
    -> );
Query OK, 0 rows affected (0.02 sec)
```

6. *Add the data shown in Figure 4 to the Item table.*

Answer:

```
mysql> INSERT INTO Item(ItemNumber, Description, OnHand, WholesalePrice, Bas
eCost, SupplierCode)
    -> VALUES
    -> ('3363', 'Baseball Cap', 110, 4.87, 6.10, 'LM'),
    -> ('3673', 'Cotton Visor', 150, 4.59, 5.74, 'LM'),
    -> ('4543', 'Crew Sweatshirt', 75, 7.29, 8.75, 'PC'),
    -> ('4583', 'Crew T-Shirt', 135, 2.81, 3.38, 'PC'),
    -> ('5923', 'Drink Holder', 75, 0.82, 1.07, 'SD'),
    -> ('6185', 'Fleece Vest', 50, 28.80, 34.50, 'PC'),
    -> ('6234', 'Foam Visor', 200, 0.79, 1.00, 'LM'),
    -> ('6345', 'Golf Shirt', 115, 10.06, 12.35, 'PC'),
    -> ('7123', 'Sports Bottle', 150, 1.04, 1.35, 'SD'),
    -> ('7934', 'Tote Bag', 225, 1.26, 1.75, 'SD'),
    -> ('8136', 'Travel Mug', 80, 2.47, 3.50, 'SD'),
    -> ('8344', 'V-Neck Pullover', 76, 13.60, 15.75, 'LM'),
    -> ('8590', 'V-Neck Vest', 65, 20.85, 25.02, 'LM'),
    -> ('9458', 'Windbreaker', 54, 15.17, 18.20, 'PC'),
    -> ('9620', 'Zip Hoodie', 90, 16.67, 20.04, 'PC') ;
Query OK, 15 rows affected (0.01 sec)
Records: 15  Duplicates: 0  Warnings: 0
```

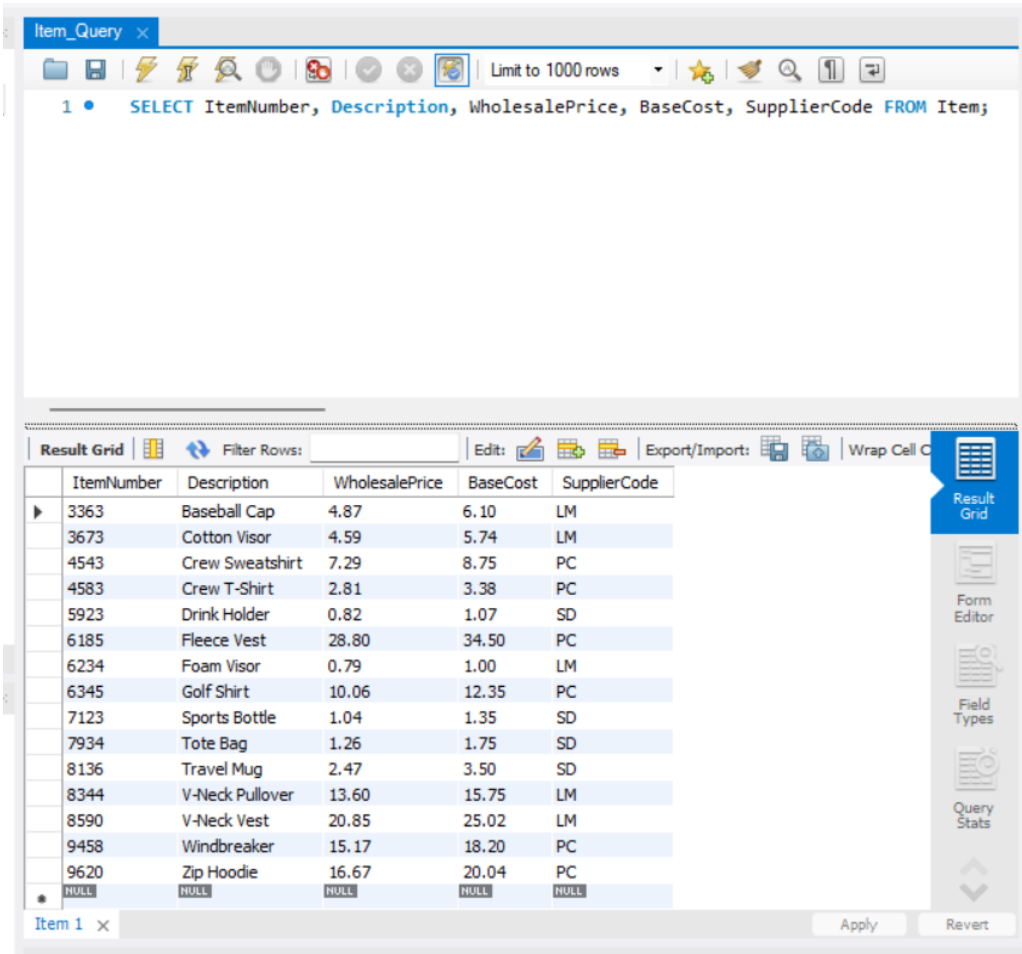
7. *Print the Item table.*

Answer:

```
mysql> SELECT * FROM Item ;
+------------+-----------------+--------+----------------+----------+--------------+
| ItemNumber | Description     | OnHand | WholesalePrice | BaseCost | SupplierCode |
+------------+-----------------+--------+----------------+----------+--------------+
| 3363       | Baseball Cap    |    110 |           4.87 |     6.10 | LM           |
| 3673       | Cotton Visor    |    150 |           4.59 |     5.74 | LM           |
| 4543       | Crew Sweatshirt |     75 |           7.29 |     8.75 | PC           |
| 4583       | Crew T-Shirt    |    135 |           2.81 |     3.38 | PC           |
| 5923       | Drink Holder    |     75 |           0.82 |     1.07 | SD           |
| 6185       | Fleece Vest     |     50 |          28.80 |    34.50 | PC           |
| 6234       | Foam Visor      |    200 |           0.79 |     1.00 | LM           |
| 6345       | Golf Shirt      |    115 |          10.06 |    12.35 | PC           |
| 7123       | Sports Bottle   |    150 |           1.04 |     1.35 | SD           |
| 7934       | Tote Bag        |    225 |           1.26 |     1.75 | SD           |
| 8136       | Travel Mug      |     80 |           2.47 |     3.50 | SD           |
| 8344       | V-Neck Pullover |     76 |          13.60 |    15.75 | LM           |
| 8590       | V-Neck Vest     |     65 |          20.85 |    25.02 | LM           |
| 9458       | Windbreaker     |     54 |          15.17 |    18.20 | PC           |
| 9620       | Zip Hoodie      |     90 |          16.67 |    20.04 | PC           |
+------------+-----------------+--------+----------------+----------+--------------+
15 rows in set (0.00 sec)
```

8. *Create a query for the Item table. Include the Item Number, Description, Wholesale Price, Base Cost, and Supplier Code. Save the query as* *<u>Item Query</u>.*

Answer:



```
SELECT ItemNumber, Description, WholesalePrice, BaseCost, SupplierCode FROM Item;
```

ItemNumber	Description	WholesalePrice	BaseCost	SupplierCode
3363	Baseball Cap	4.87	6.10	LM
3673	Cotton Visor	4.59	5.74	LM
4543	Crew Sweatshirt	7.29	8.75	PC
4583	Crew T-Shirt	2.81	3.38	PC
5923	Drink Holder	0.82	1.07	SD
6185	Fleece Vest	28.80	34.50	PC
6234	Foam Visor	0.79	1.00	LM
6345	Golf Shirt	10.06	12.35	PC
7123	Sports Bottle	1.04	1.35	SD
7934	Tote Bag	1.26	1.75	SD
8136	Travel Mug	2.47	3.50	SD
8344	V-Neck Pullover	13.60	15.75	LM
8590	V-Neck Vest	20.85	25.02	LM
9458	Windbreaker	15.17	18.20	PC
9620	Zip Hoodie	16.67	20.04	PC
NULL	NULL	NULL	NULL	NULL

9. *Create a simple form for the Item table. Use the name* *<u>Item Form</u>* *for the form.*

Answer:

10. *Drop the Supplier table.*

Answer:

```
mysql> ALTER TABLE Item DROP FOREIGN KEY item_ibfk_1;
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DROP TABLE Supplier;
Query OK, 0 rows affected (0.01 sec)
```

.

Structure of Supplier Table

FIELD NAME	*DATA TYPE*	*FIELD SIZE*	*PRIMARY KEY?*	*DESCRIPTION*
Supplier Code	*Short Text*	*2*	*Yes*	*Supplier Code (Primary Key)*
Supplier Name	*Short Text*	*30*		*Supplier Name*
Telephone Number	*Short Text*	*12*		*Supplier Telephone Number*

Figure 1

Data for Supplier Table

Supplier Code	*Supplier Name*	*Telephone Number*
LM	*Last Merchandisers*	*803-555-7641*
PC	*Pratt Clothing*	*517-555-3853*
SD	*Scryps Distributors*	*610-555-8741*

Figure 2

Structure of Item Table

FIELD NAME	*DATA TYPE*	*FIELD SIZE*	*PRIMARY KEY?*	*DESCRIPTION*
Item Number	*Short Text*	*4*	*Yes*	*Item Number (Primary Key)*
Description	*Short Text*	*20*		*Description of Item*
On Hand	*Number*			*On-Hand Item*
Wholesale Price	*Currency*			*Wholesale Price (Caption)*
Base Cost	*Currency*			*Base Cost for Item*
Supplier Code	*Short Text*	*2*		*Supplier Code*

Figure 3

Data for Item Table

Item Number	*Description*	*On Hand*	*Wholesale Price*	*Base Cost*	*Supplier Code*
3363	*Baseball Cap*	*110*	*$4.87*	*$6.10*	*LM*
3673	*Cotton Visor*	*150*	*$4.59*	*$5.74*	*LM*
4543	*Crew Sweatshirt*	*75*	*$7.29*	*$8.75*	*PC*
4583	*Crew T-Shirt*	*135*	*$2.81*	*$3.38*	*PC*
5923	*Drink Holder*	*75*	*$0.82*	*$1.07*	*SD*
6185	*Fleece Vest*	*50*	*$28.80*	*$34.50*	*PC*
6234	*Foam Visor*	*200*	*$0.79*	*$1.00*	*LM*
6345	*Golf Shirt*	*115*	*$10.06*	*$12.35*	*PC*
7123	*Sports Bottle*	*150*	*$1.04*	*$1.35*	*SD*
7934	*Tote Bag*	*225*	*$1.26*	*$1.75*	*SD*
8136	*Travel Mug*	*80*	*$2.47*	*$3.50*	*SD*
8344	*V-Neck Pullover*	*76*	*$13.60*	*$15.75*	*LM*
8590	*V-Neck Vest*	*65*	*$20.85*	*$25.02*	*LM*
9458	*Windbreaker*	*54*	*$15.17*	*$18.20*	*PC*
9620	*Zip Hoodie*	*90*	*$16.67*	*$20.04*	*PC*

Figure 4